

Table 1: Few-shot only (different prompts and lengths than the zero-shot table—do not mix rows). Falcon-Mamba BABILong ablation results. Accuracy is exact-match (%) averaged over QA1–QA3 (50 per cell). *No cache, few-shot*: reasoning-curves approximate next-token/prefix baseline, QA1–QA3, stored as `metrics/fewshot_babilong/no_cache_baseline.json`. *StateCache rows*: `cache_acc` from `metrics/fewshot_babilong/{run_id}.json` (`eval_babilong_paper_scaffold_checkpoint`, paper-style few-shot + [CACHE]).

Method	0k	1k	2k	4k	8k	16k	Avg.
No cache, few-shot	52.0	36.7	28.0	24.7	5.3	0.0	24.4
Loss-selected, $k = 1$, L62	28.0	33.3	34.0	16.7	18.7	18.0	24.8
Loss-selected, $k = 4$, L62	29.3	34.7	36.7	14.7	22.7	18.7	26.1
Loss-selected, $k = 8$, L62	27.3	40.0	37.3	18.7	17.3	20.7	26.9
Loss-selected, $k = 16$, L62	28.7	33.3	34.0	12.0	20.7	14.0	23.8
Loss-selected, $k = 32$, L62	32.0	40.0	42.0	18.0	17.3	20.7	28.3
Loss-selected, $k = 64$, L62	31.3	34.0	38.7	21.3	18.7	16.7	26.8
Loss-selected, $k = 4$, L62, train up to 8k	25.3	30.7	30.0	34.0	33.3	24.0	29.6
Loss-selected, $k = 4$, L62, no gate	32.0	36.0	37.3	20.7	18.7	24.0	28.1
Loss-selected, $k = 4$, L62, full loss	27.3	18.0	15.3	15.3	15.3	16.0	17.9
Loss-selected, $k = 4$, L32	48.7	43.3	36.7	27.3	30.7	26.0	35.4
Loss-selected, $k = 4$, L2	64.7	54.7	51.3	45.3	31.3	28.0	45.9
Uniform interval, $k = 4$, L62	31.3	31.3	38.7	19.3	22.0	18.0	26.8
Random, $k = 4$, L62	27.3	21.3	21.3	18.0	22.0	13.3	20.6
Regex/entity, L62	28.7	28.7	36.0	18.0	13.3	21.3	24.3